Deena A.C

Firmware Engineer

Tamil Nadu, India | 7deena19@gmail.com | +91 9043291234
linkedin.com/in/acdeena1994 | acdeena1994 | Portfolio

Summary

Firmware Engineer (B.Tech CSE–IoT, CGPA 8.53) with hands-on experience in **embedded C/C++** firmware development on **ESP32**, sensor interfacing over **I2C**, **SPI**, and **UART**, **FreeRTOS** multitasking, and **Edge AI** integration. Experienced in designing, programming, debugging, and testing embedded firmware subsystems, with a strong foundation in real-time control systems and computer networking. Published in **IEEE** and **Springer** conferences; 2nd prize winner at the **IEEE RAS Hackathon**. Eager to learn firmware engineering, and open to relocation and international opportunities.

Skills

Languages: C, C++ (Embedded C), Python (Basics)
Microcontrollers & RTOS: ESP32, ESP8266, Bare-Metal Programming, Arduino, FreeRTOS (multi-core task)
Protocols & Interfaces: I2C, SPI, UART, MQTT, HTTP/REST, WebSockets, TCP/IP
Sensors & Hardware: PZEM-004T, ACS712, LDR, PIR, SSD1306 OLED, Relays, KiCad (Beginner)
Edge AI & Software: YOLO (Ultralytics), OpenCV, Machine Learning, Flask, Socket.IO, SQLite, JSON
IoT Cloud & Tools: Blynk, ThingSpeak, Arduino IoT Cloud, Arduino IDE, Git/GitHub
Core Expertise: Firmware Development, System Architecture, Energy Optimization, NILM

Projects

Adaptive Non-Intrusive Load Monitoring (NILM) System *Jul 2025 – May 2026*
Tech Stack: **ESP32**, **C++/Arduino**, **PZEM-004T**, **SSD1306 OLED (I2C)**, **Wi-Fi/HTTP**, **Python**, **Flask**, **Socket.IO**, **JSON**, **Machine Learning**

- **Data Acquisition:** The **ESP32** edge node polls the **PZEM-004T** sensor every second to capture voltage, current, power, energy, frequency, and power factor, displays readings on the **OLED**, and detects significant power events using a **20 W threshold**.
- **Data Transmission & Processing:** Sends structured **JSON** payloads via **HTTP POST** (with **API-key authentication**) to a **Flask** backend, which applies **transition detection** and **multi-parameter signature matching** to identify individual appliances from aggregate data.
- **Adaptive Learning & Visualization:** Streams results to a real-time **Socket.IO** dashboard where users give feedback on ambiguous events, continuously refining the model and improving **disaggregation accuracy** over time.

Intelligent Building Automation System – YOLO Computer Vision + ESP32 IoT *Jul 2025 – Dec 2025*
Tech Stack: **ESP32**, **IP Camera/Webcam**, **Python**, **Flask**, **Ultralytics YOLO**, **OpenCV**, **SQLite (WAL mode)**, **Arduino (C++)**, **HTTP REST**, **WebSockets**, **TCP/IP**

- **Vision-Based Detection:** A **Python** backend captures live frames from an **IP camera**; **YOLO** detects people (class 0) and calculates real-time **headcount**.
- **Automated Control Logic:** In auto mode, headcount is compared against **per-device thresholds** (e.g., Light 1 at 1 person, Fan 2 at 3); all appliances turn off at zero occupancy unless **manual override** is active. **Flask** auto-discovers the **ESP32** IP and sends **HTTP POST JSON** commands to toggle **relays**.
- **Real-Time Updates: Flask-SocketIO** pushes headcounts, device states, and capacity alerts (limit = 5) instantly to the web dashboard; **SQLite (WAL)** persists logs for concurrent read/write.

Hybrid Energy Harvesting System for Smart Street Lighting *Dec 2024 – Mar 2025*
Tech Stack: **ESP32**, **FreeRTOS**, **Arduino (C++)**, **Adafruit SSD1306 OLED**, **ArduinoJson**, **HTTPClient**, **Flask (Python)**

- **System Init & Multitasking:** Initialized an **I2C OLED** (128×64), configured **GPIO** for **LDR**, **PIR**, **ACS712** current, and battery/solar voltage sensors, plus actuators (LED, relay); connected Wi-Fi and spawned two **FreeRTOS** tasks pinned to separate cores via `xTaskCreatePinnedToCore`, leaving `loop()` empty.
- **Sensor Task (Core 0):** Runs every 1 s – acquires analog sensor data, dynamically switches OLED animations (Robo Eye at $\leq$1.0 V solar, Charging Battery above), drives the LED via an **LDR threshold** and the relay via **PIR motion** with a **non-blocking 5 s millis() timer**.
- **HTTP Task (Core 1):** Runs every 5 s – monitors and reconnects Wi-Fi via `checkWiFiConnection()`, serializes telemetry into **JSON** with **ArduinoJson**, and sends **HTTP POST** to a **Flask** endpoint for real-time dashboard updates.

Experience

Electric Vehicle Technology – Kodacy *Jun 2026 – Jul 2026 (Remote)*

- Analyzed core EV subsystems, including electric motors, power electronics (converters/inverters), Battery Management Systems (BMS), and transmission controllers.
- Evaluated regenerative braking, charging protocols, and e-differentials through practical simulations to understand energy flow and performance optimization.

Patents & Publications

- **Indian Patent Application (IPO) – *Vision Track Systems*** *2026*
  Filed Apr 2026 and published Jun 2026; request for examination pending.
- **IEEE GCWCN 2025:** *Adaptive Non-Intrusive Load Monitoring with Human-in-the-Loop Machine Learning for Appliance Disaggregation.*
- **Springer ICICCD 2025:** *Intelligent Building Automation System using YOLOv8 and ESP32 for Energy-Efficient Occupancy Detection.*
- **IEEE ICCSC 2024:** *Experimental Investigation for Hybrid Energy Harvesting System for Remote Monitoring Applications.*

Certifications

Embedded Systems with Arduino – GeeksforGeeks View Certificate
Introduction to C – Sololearn View Certificate
Introduction to C++ – Sololearn View Certificate
C++ Intermediate – Sololearn View Certificate

Achievements

- **IEEE RAS Hackathon** – 2nd Prize for technical innovation
- **IEEE Conference Presenter** and **Springer Conference Publication**
- **Ignite India** – Wadhwani Foundation

Education

B.Tech – Computer Science and Engineering (IoT) *Sep 2022 – Apr 2026*
Kalasalingam Academy of Research and Education, Virudhunagar, Tamil Nadu CGPA: **8.53**

Higher Secondary School *2021 – 2022*
Fusco's Matriculation Higher Secondary School, Madurai, Tamil Nadu Percentage: **81%**